

HARIPRIYA PULYASSARY

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EDUCATION

AUG 2027 **PhD Student – Operations Research**
(Expected) **(Cornell University)**

Advisors: Prof. David Shmoys & Prof. Manxi Wu

Research focus: Design of optimization methods to improve the sustainability of transportation and logistics networks, particularly in the context of electric vehicles

Selected Coursework: Mathematical Programming I, Applied Stochastic Processes, Scheduling Theory, Analysis of Algorithms, Statistical Principles, Foundations of Reinforcement Learning.

AUG 2022 **Masters of Mathematics in Combinatorics & Optimization**
(University of Waterloo)

Advisor: Prof. Chaitanya Swamy

Research focus: Algorithm design for facility location and clustering problems in ordinal settings (i.e. limited information regarding the underlying distances is known).

Coursework: Combinatorial Optimization, Semidefinite Programming

APR 2021 **Bachelors in Computer Science, Co-op**
(University of Waterloo)

Joint Honours in Combinatorics & Optimization,
Minor in Statistics

PUBLICATIONS

- 2026 Adaptive sampling algorithms for minimum-norm k -clustering
H. Pulyassary and C. Swamy, *European Symposium on Algorithms (ESA)* 2026.
- 2025 Network Flow Problems with Electric Vehicles
H. Pulyassary, K. Kollias, A. Schilds, D. Shmoys, M. Wu, *Mathematical Programming B, IPCO* 2024
- 2025 Constant-factor distortion mechanisms for k -committee elections
H. Pulyassary and C. Swamy, *AAAI* 2025 (Oral paper)
- 2024 Market Design for Capacity Sharing in Networks
S. Amin, P. Jalliet, H. Pulyassary, M. Wu, *WINE* 2024
- 2024 Digital Transformation in Transportation Systems: Navigating User Behavior and System Efficiency
H. Pulyassary and M. Wu, *INFORMS 2024 tutORial*
- 2023 Capacity Allocation and Pricing of High Occupancy Toll Lane Systems with Heterogeneous Travelers
H. Pulyassary, R. Yang, Z. Zhang, M. Wu, *CDC* 2023
- 2021 The aggregation closure is polyhedral for packing and covering integer programs.
K. Pashkovich, L. Poirrier, and H. Pulyassary, *Mathematical Programming*

OTHER ARTICLES

- 2026 Infrastructure-design for Heavy-duty Electric Vehicles
H. Pulyassary, D. Shmoys, and M. Wu, working paper.
- 2021 On the Randomized Metric Distortion Conjecture
H. Pulyassary and C. Swamy, on ArXiv.

INTERNSHIPS

AUG 2026	PhD Scientist Intern
MAY 2026	UBER TECHNOLOGIES EATER, PRICING & INCENTIVES
SEPT 2025	PhD Scientist Intern
DEC 2025	UBER TECHNOLOGIES EATER, PRICING & INCENTIVES
MAY 2020	Undergraduate Research Assistant
AUG 2020	UNIVERSITY OF WATERLOO DEPT. COMBINATORICS & OPTIMIZATION
JAN 2020	Software and Data Engineering Intern
APR 2020	NATIONAL BANK OF CANADA
MAY 2019	Undergraduate Research Assistant
AUG 2019	UNIVERSITY OF WATERLOO DEPT. COMBINATORICS & OPTIMIZATION
MAY 2018	Data Science and Solutions Intern
AUG 2018	ZIGHRA
MAY 2017	Data Science Intern
AUG 2017	ZIGHRA

TEACHING EXPERIENCE

Course Instructor

SUMMER 2025	<i>Co-instructor of ENGRI 1101 Engineering Operations: Data Science and Decision Making</i>	Cornell University
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Teaching Assistant

SPRING 2026	<i>ORIE 4350 Introduction to Game Theory</i>	Cornell University
SPRING 2025	<i>ORIE 3310 Optimization II</i>	Cornell University
SPRING 2024	<i>ORIE 4742 Info Theory, Probabilistic Modeling, and Deep Learning</i>	Cornell University
FALL 2022	<i>ORIE 4350 Introduction to Game Theory</i>	Cornell University
SUMMER 2022	<i>CO 250 Introduction to Optimization – Head TA</i> • Received the Outstanding TA award	University of Waterloo
WINTER 2022	<i>CO 353 Computational Discrete Optimization</i> • Nominated for the Outstanding TA award	University of Waterloo
FALL 2021	<i>CO 255 Introduction to Optimization (Advanced)</i>	University of Waterloo
SUMMER 2021	<i>CO 250 Introduction to Optimization</i>	University of Waterloo

OTHER ACTIVITIES

2023-2025	PhD colloquium & Discrete Optimization Reading Group co-organizer (Cornell ORIE)
2024 & 2025	Co-organizer of Cornell-Columbia-NYU joint student colloquium
2023-2025	Cornell INFORMS student chapter: Secretary
2022-2023	First-year Colloquium co-organizer (Cornell ORIE)